

## TECHNICAL AND FINANCIAL OFFER

### Steel-framed sandwich-panel roof over the existing roof slab Administration Building, Regatta Tourist Village, Tripoli

|           |                                                  |              |                                       |
|-----------|--------------------------------------------------|--------------|---------------------------------------|
| Offer No. | RTV-2026-01                                      | Date         | 29 September 2026                     |
| To        | Regatta Tourist Village - Administration         | Valid until  | 14 October 2026                       |
| From      | Eng. MALEK ABOZRAIG,<br>malek.abozraig@gmail.com | Design basis | Drawings RTV-ST-S00 to S06,<br>Rev 6a |

Dear Sirs,

Further to the survey of the existing administration building and the structural design completed for it, we are pleased to submit our offer for the supply, fabrication and erection of a steel roof structure with insulated sandwich-panel cladding over the existing roof slab, as described below and shown on the attached drawings. The offer is a lump sum covering all materials, workshop fabrication, transport, site works and erection listed in the scope, delivered as a complete and weather-tight roof.

#### 1. The structure

The new roof is a light steel post-and-beam structure standing on the existing 300 mm roof slab, with one steel column on each of the 27 existing concrete columns so that all loads pass straight into the building's own structure. Level primary beams run along the column rows; sloping rafters span between them and carry cold-formed purlins and 50 mm insulated PIR sandwich panels on a single roof plane falling 6 % to the north, where two box gutters and four downpipes on the north facade take the rainwater away. Lateral stability is provided by eight X-braced wall bays and a braced roof plane, designed for the Tripoli wind and seismic conditions. The stair well and the lift shaft remain open, with upstands and flashings on all sides. The roofed area is 439 m<sup>2</sup> over an L-shaped footprint of about 27.8 by 20.3 m; the clear height under the steel is 3.0 m at the low north eave and about 4.3 m at the south.

| Element              | Specification                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Columns              | 27 no. HEA 140, S275, one over each existing concrete column, plus one wind post                                           |
| Primary beams        | IPE 300 (one 9.8 m beam IPE 330), S275, level, bolted cap plates on the columns                                            |
| Rafters and trimmers | IPE 240, S275, single roof plane at 6 % falling north, bolted fin plates                                                   |
| Purlins and girts    | Z 200 x 2.0 cold-formed, S350GD galvanised, at 1.5 m centres                                                               |
| Bracing              | L 70 x 7 wall diagonals in 8 bays; M24 roof rods in 13 panels                                                              |
| Bases                | 4 post-installed dia 16 rebars per column into the existing column head, grouted shear keys, 20 mm plates, 25 mm grout bed |
| Roof cladding        | PIR sandwich panels 50 mm, colour-coated steel both faces, with ridge, eave and verge flashings and well upstands          |
| Rainwater            | Two box gutters 150 x 100 with overflows, four 100 mm downpipes on the north facade                                        |
| Steel weight         | about 22 tonnes, all steel hot-dip galvanised or shop-primed and painted as agreed                                         |

## 2. Scope of supply and works

- Detailed fabrication drawings and the bill of materials based on the approved design (Rev 6a).
- Supply of all structural steel sections, plates, bolts, rebars, anchors and cold-formed purlins and girts.
- Workshop fabrication, surface protection, transport to site and offloading.
- Site survey of the slab top, scanning of the 27 column heads, drilling and injection of the anchors, coring and grouting of the shear keys, and proof tests on three bars.
- Erection of columns, beams, rafters, bracing and purlins with all connections, including the temporary bracing needed during erection.
- Supply and fixing of the roof sandwich panels with all flashings, well upstands, sealants and fasteners.
- Supply and fixing of gutters, downpipes and the associated brackets and flashings.
- Site management, safety measures, cleaning of the site and handover with as-built drawings.

## 3. Not included

- Wall cladding panels and their girts (to be offered separately once the product is selected).
- Verification of the existing building (as-built drawings, slab-top survey drawing, adequacy statement of the existing columns and foundations) and any strengthening it may require.
- Building permits, authority fees, independent checking and any works to the existing slab or facade beyond the anchors, keys and downpipe fixings.
- Electrical, mechanical, lighting, ceilings, floors and finishes under the new roof.

## 4. Programme

The works are planned over about three weeks from receipt of the advance payment, subject to site access and to the scan results at the column heads.

| Week | Activity                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 1    | Fabrication drawings approved; slab survey and column-head scans; steel and panel procurement; workshop fabrication starts |
| 2    | Fabrication completed and delivered; anchors drilled and injected, shear keys cored and grouted; columns and beams erected |
| 3    | Rafters, bracing and purlins completed; roof panels, flashings, gutters and downpipes fixed; cleaning and handover         |

## 5. Price

|                                                                                                                                                                                           |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Lump-sum price for the complete scope in section 2</b>                                                                                                                                 | <b>LYD 404,000</b> |
| Libyan Dinars, fixed for the validity period, inclusive of materials, fabrication, transport, site works, erection and supervision. Taxes and fees, if any, are for the client's account. |                    |

## 6. Payment terms

| Stage                                              | Share | Amount, LYD |
|----------------------------------------------------|-------|-------------|
| Advance on signing the order; starts the programme | 50 %  | 202,000     |
| On start of erection on site                       | 25 %  | 101,000     |
| On completion and handover                         | 25 %  | 101,000     |

## 7. Conditions

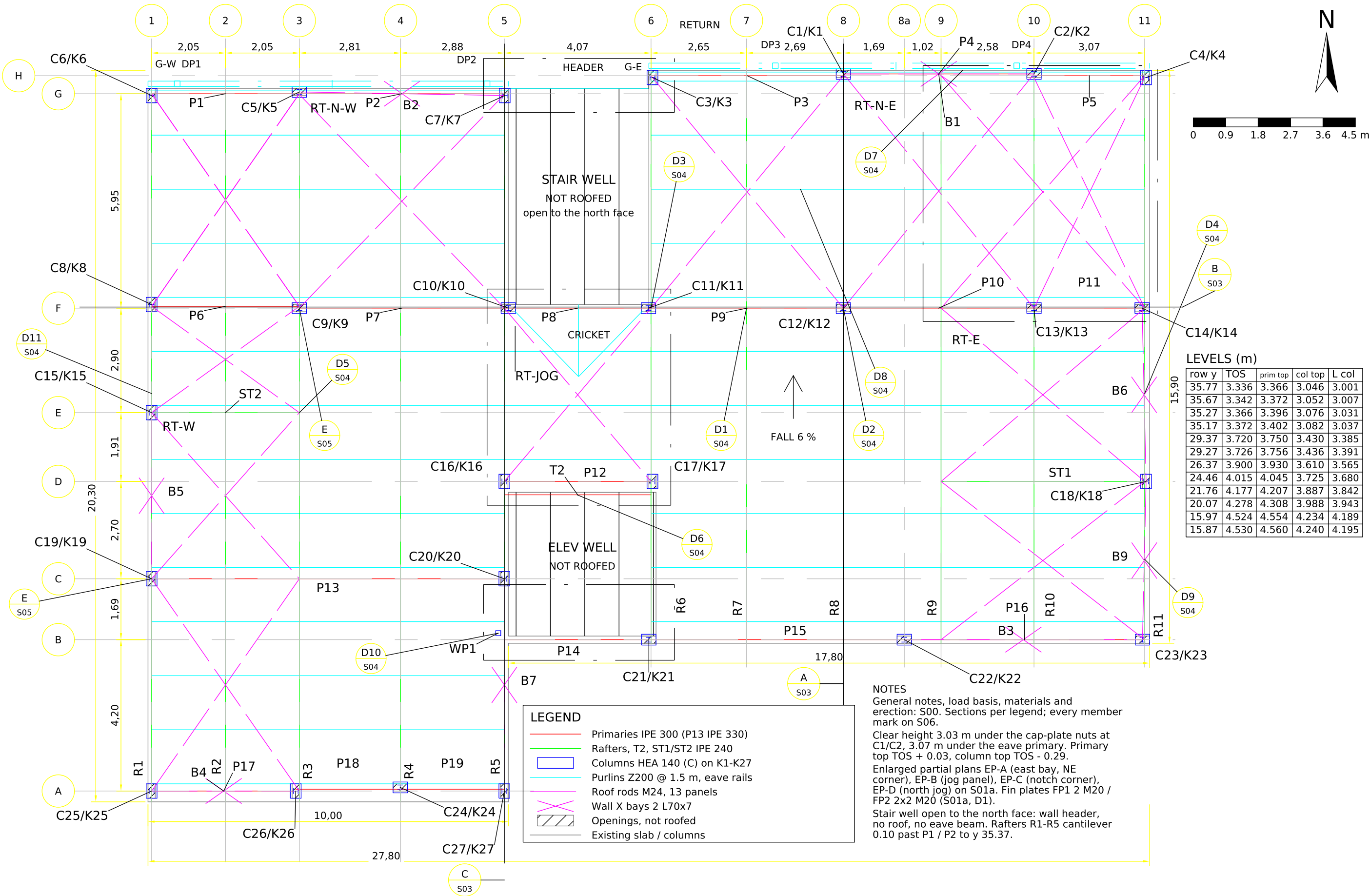
- Validity: this offer is valid for 15 days from its date, until 14 October 2026, because of the movement of steel prices.
- Warranty: 12 months from handover on the steel structure, panel fixings and workmanship; panel and coating warranties as per the manufacturer.
- The design and drawings are based on the assumptions listed on drawing S00. The client provides as-built information on the existing building and free access to the roof and to the north facade for the downpipes; anything found at the column heads that differs from the assumptions (bar positions, concrete quality, slab thickness) is dealt with by the fallback details on drawing S05 and, if it changes the scope, by a written variation before work continues.
- The client confirms that the eight braced wall bays shown on drawing S01 remain free of doors and openings.
- Materials: structural steel S275 to EN 10025, bolts 8.8, cold-formed purlins S350GD, PIR panels with a manufacturer's certificate; anchor system with a European Technical Assessment for post-installed rebar.
- Design standards: Eurocodes EN 1990, 1991, 1993 and 1998 with parameters for Tripoli, as recorded in the design report.
- Any change to the roof plane, the openings, the cladding type or the scope after the order is priced separately.
- The offer excludes anything not expressly stated in section 2.

## 8. Attachments

- Attachment A: Drawing RTV-ST-S01, roof framing plan (A3).
- Attachment B: Drawing RTV-ST-S03, typical sections (A3).
- Attachment C: 3D view of the structure.
- The complete set of drawings RTV-ST-S00 to S06 (A2, Rev 6a) is available on request and forms part of the order.

We remain at your disposal for any clarification and look forward to working with you on this project.

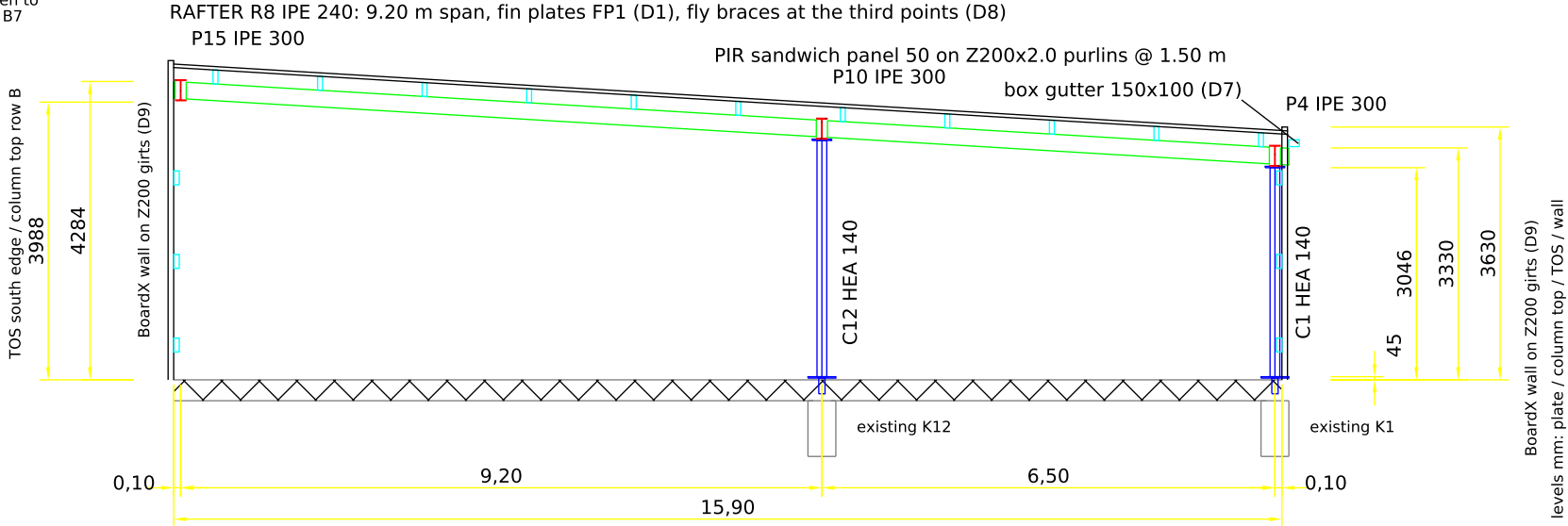
| For the offer                                                  | For the client (acceptance)                   |
|----------------------------------------------------------------|-----------------------------------------------|
| Eng. MALEK ABOZRAIG<br>malek.abozraig@gmail.com<br>Date: ..... | Name: .....<br>Position: .....<br>Date: ..... |



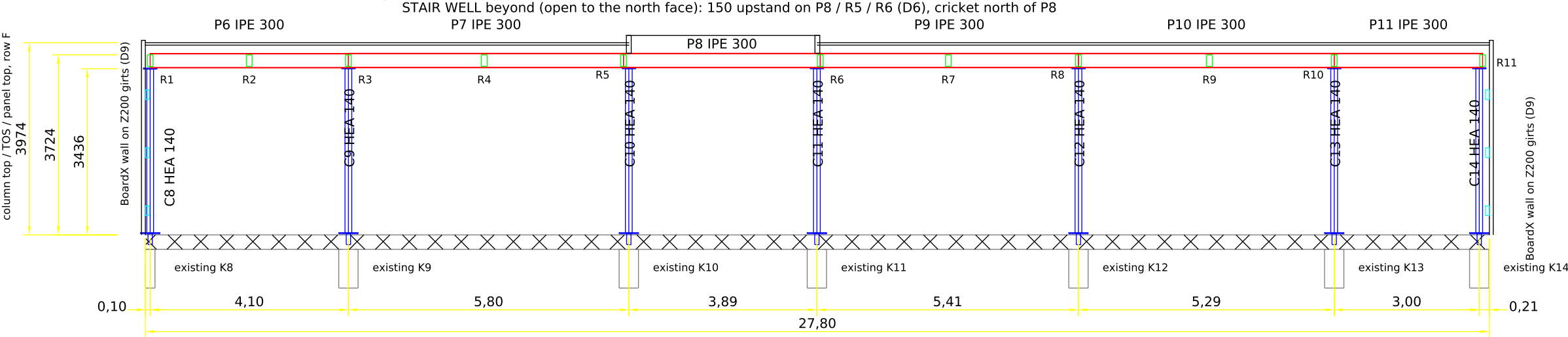
|                                                                                                                                                                                                                                                                              |                                                                                                                                                                                  |           |            |                                                                                                                                                                        |                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| PROJECT<br>Steel roof - Administration Building, Regatta Tourist Village, Tripoli<br><br>ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS<br><br>Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m | SHEET TITLE<br>ROOF FRAMING PLAN<br><br>Scale 1:1 in model space (m); print 1:200 on A1<br><br>Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9 | REVISIONS |            | CURRENT REVISION<br>Rev 6a superstructure<br>Rev 8 bases / bracing Rev 5a<br><br>DATE 2026-09-29<br><br>Drawn: ..... Designed: .....<br>Checked: ..... Approved: ..... | DRAWING No.<br>RTV-ST-S01<br><br><b>S01</b> |
|                                                                                                                                                                                                                                                                              |                                                                                                                                                                                  | 3         | 2026-09-29 | Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a                                                                                                             |                                             |
|                                                                                                                                                                                                                                                                              |                                                                                                                                                                                  | 2         | 2026-09-28 | Design Rev 3 / bases Rev 4b (critique C1-C15)                                                                                                                          |                                             |
|                                                                                                                                                                                                                                                                              |                                                                                                                                                                                  | 1         | 2026-09-27 | First issue: design Rev 2, bases Rev 3                                                                                                                                 |                                             |

SECTION NOTES  
A-A: north on the right; rafter R8 9.20 m span; C1 and C12 in section; gutter G-E at the north eave. Clear height 3.03 m under the cap-plate nuts at C1/C2, 3.07 m under the eave primary P4.  
B-B: east on the right; rafters R1-R11 beyond shown dashed; stair well beyond open to the north face. C-C: north on the right; R5 ends at y 35.37 (west north edge); bay B7 C27-C20; WP1 beyond.

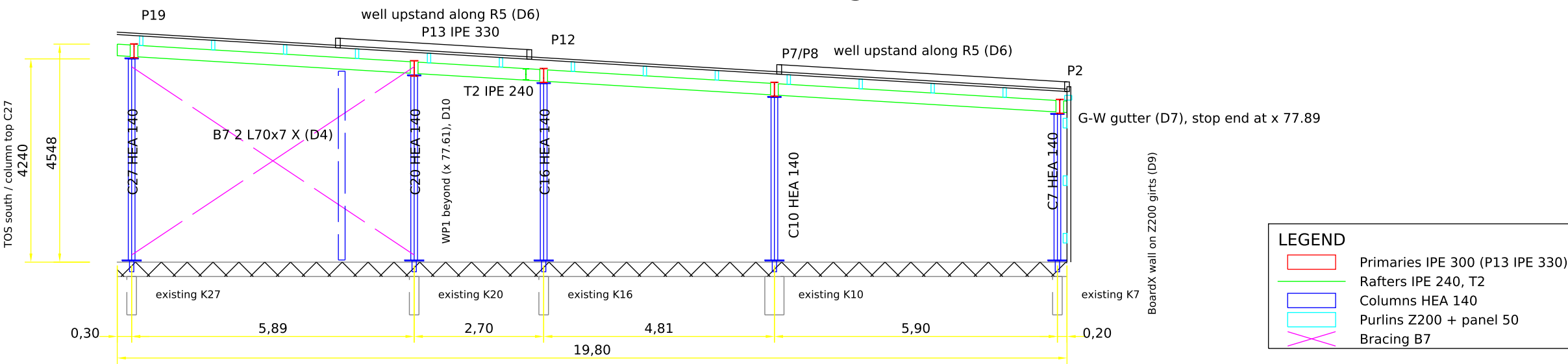
SECTION A-A rafter line 8 (x 87.19), looking west



SECTION B-B primary row F (y 29.27), looking north



SECTION C-C braced line 5 (x 77.78), looking west



- LEGEND
- Primaries IPE 300 (P13 IPE 330)
  - Rafters IPE 240, T2
  - Columns HEA 140
  - Purlins Z200 + panel 50
  - Bracing B7

PROJECT  
Steel roof - Administration Building, Regatta Tourist Village, Tripoli  
ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS  
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m

SHEET TITLE  
TYPICAL SECTIONS A-A, B-B, C-C  
Sections 1:1 in model space (m); levels in mm; print 1:100 on A1  
Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9

| REVISIONS |            |                                                            |  |
|-----------|------------|------------------------------------------------------------|--|
| 3         | 2026-09-29 | Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a |  |
| 2         | 2026-09-28 | Design Rev 3 / bases Rev 4b (critique C1-C15)              |  |
| 1         | 2026-09-27 | First issue: design Rev 2, bases Rev 3                     |  |

CURRENT REVISION  
Rev 6a superstructure  
Rev 8 bases / bracing Rev 5a  
DATE 2026-09-29  
Drawn: ..... Designed: .....  
Checked: ..... Approved: .....

DRAWING No.  
RTV-ST-S03  
S03

NE isometric, roof panel hidden: purlins, rafters and roof rods - Alternative C, Rev 3

